

Depth for Serial, Width for Parallel: How Transformers Allocate Layers and Attention Heads to a Task’s Computational Structure

(author list omitted in draft)

August 12, 2026

Abstract

Expressivity theory tells us what a transformer *can* compute: fixed depth bounds the number of *sequential* steps available, while heads supply *parallel* capacity. We ask the complementary empirical question: given a trained model and a task, how much of each resource does it actually *allocate*? We characterise every task by two structural coordinates of its dependency graph—serial depth D (the critical path) and parallel width W (the largest set of mutually independent intermediate values)—and every prediction by two allocation measures read off the network’s active-head landscape: recruited width $\hat{W}$ (active heads per layer) and recruited depth $\hat{D}$ (the active frontier). Our organising hypothesis is a *resource-matching law*: $\hat{D}$ should track D and $\hat{W}$ should track W , with negligible cross-terms, at exchange rates c (layers per serial step) and γ (heads per parallel item) that in turn predict a capability ceiling $D_{\max} \approx L_{\text{eff}}/c$. We test this with a factorial suite that varies D and W independently under a length control. On Llama-3-70B the width half holds cleanly—parallel width recruits heads ($r=+0.81$ length-matched; grid fit $\hat{W} \approx 32.3 - 0.14 D + 0.25 W$) while serial depth recruits none—and the length control proves necessary, since the uncontrolled ladder trends the wrong way. The depth half is null on this model, which the framework explains as the expected *sub-ceiling* regime: a 70B idles $\sim 50\%$ of its heads, so a six-step task never approaches its serial budget. In lower-reserve models the depth axis does appear (Llama-3-8B: $r(\text{hops, frontier})=+0.96$; pooled $r=+0.70$ across the largest GPT-2/Pythia models), and recruitment overall scales with model capacity. We then show how architecture sets the two budgets—depth L , heads H , KV groups n_{kv} under grouped-query attention, experts under mixture-of-experts, and the aspect ratio L/H —and derive falsifiable cross-architecture predictions. Finally we delimit what active-node geometry does and does not reveal about circuits: it constrains a circuit’s support, width and depth-extent, but not its wiring; combined with a known task graph it identifies the *embedding* of that graph into the layer $\times$ head lattice, which is the sense in which task structure can be read from activation structure. Complete data (84 measurements) and an experimental roadmap are included.

1 Introduction

Residual transformers expose, at every layer, two routes for information: a *compute path* through the attention heads and the feed-forward (FFN) block, and an *identity residual path* that bypasses them. A large body of work establishes that the compute path is heavily underused—most attention heads can be pruned with little loss (Michel et al., 2019; Voita et al., 2019), deep layers are frequently redundant (Men et al., 2024; Gromov et al., 2024), and harder inputs deserve more computation (Graves, 2016; Schuster et al., 2022). What is missing is a *graded, per-task* account: not “which heads are removable in general,” but **how the set of working heads grows as a task becomes harder, and how that growth is bounded by model capacity.**

A parallel theoretical literature makes the resource structure precise. Fixed-depth log-precision transformers occupy a shallow parallel class with a *bounded number of sequential steps* (Merrill & Sabharwal, 2023); chain-of-thought tokens are how a model buys serial steps it cannot fit in depth (Merrill & Sabharwal, 2024; Li et al., 2024; Feng et al., 2023); and the dichotomy is explicit in expressivity results showing that parallel-style problems are solvable by one wide layer while k -hop and pointer-chasing problems *provably require depth* (Sanford et al., 2023, 2024). These are statements about what an architecture *can* compute in the worst case. They leave open the empirical question this paper addresses:

*Given a **trained** model and a specific task, how much serial (depth) and parallel (head) resource does it actually allocate—and does the allocation match the task’s own serial and parallel structure?*

We answer by giving both sides of that sentence a measurable definition. On the task side, every task carries two structural coordinates of its dependency graph: *serial depth* D (the critical path) and *parallel width* W (the largest set of mutually independent intermediate values). On the network side, every prediction induces an active-head set whose geometry gives *recruited depth* $\hat{D}$ (how deep the active region reaches) and *recruited width* $\hat{W}$ (how many heads per layer are engaged). The organising hypothesis is a **resource-matching law**: $\hat{D}$ tracks D , $\hat{W}$ tracks W , and the cross-terms vanish (Section 3).

Two consequences make this more than a restatement. First, the diagonal coefficients are *exchange rates with units*— c layers per serial step, γ heads per parallel item—and c predicts a capability ceiling $D_{\max} \approx L_{\text{eff}}/c$ from architecture alone. Second, the budgets are set by architectural choices (L , H , KV groups, experts, aspect ratio L/H), so the law generates falsifiable cross-architecture predictions (Section 6).

Contributions.

1. **A framework** that makes task complexity and network allocation commensurable: definitions of (D, W) and $(\hat{D}, \hat{W})$, the resource-matching law with its exchange rates, and the capacity ceiling it implies (Section 3).
2. **A controlled (D, W) factorial** that varies serial depth and parallel width independently under a length control, establishing on Llama-3-70B that *parallel width recruits heads and serial depth does not* (Section 5.5).
3. **Supporting allocation results**: demand-graded recruitment, its dilation with model scale, and frontier extension with reasoning depth in lower-reserve models (Sections 5.1–5.3).
4. **An architectural account** of how L , H , n_{kv} , experts and aspect ratio set the two budgets, with cross-architecture predictions (Section 6).
5. **A delimitation of what active-node geometry says about circuits**—support, width and depth-extent, but not wiring—and the sense in which task structure is recoverable from activation structure (Section 7).
6. The complete 84-measurement dataset (Appendix A) and an experimental roadmap (Section 9).

2 Related work

Attention-head and layer redundancy. Michel et al. (2019) showed that a large fraction of attention heads can be pruned at test time with negligible loss, and that importance is highly uneven

across heads. Voita et al. (2019) reached a similar conclusion for encoder self-attention, identifying a small set of specialised heads (positional, syntactic, rare-token) that do most of the work while the rest are removable. At the layer level, Men et al. (2024) introduce a *Block Influence* score—the degree to which a block changes the hidden state—and find many deep layers contribute little; Gromov et al. (2024) likewise prune large contiguous spans of late layers with small degradation. Fan et al. (2020) (LayerDrop) and Sajjad et al. (2023) study structured layer dropping during and after training. *All of this work treats redundancy as a (largely) static property of a trained network. Our contribution is to make redundancy a measured, monotone function of task demand—the dead set is not fixed but is drawn down as the task gets harder.*

Conditional and adaptive computation. Graves (2016) introduced Adaptive Computation Time, letting a recurrent model learn how many steps to spend per input. The idea was carried into transformers by Universal Transformers (Dehghani et al., 2019) and by depth-adaptive / early-exit methods (Elbayad et al., 2020; Xin et al., 2020; Schuster et al., 2022), which halt computation once a per-token confidence criterion is met. Raposo et al. (2024) (Mixture-of-Depths) route only some tokens through each block. *These methods adapt essentially one scalar—how deep to go. We provide evidence for a second, orthogonal axis the model already uses internally: how wide (how many heads) to engage at each layer, as a graded function of demand.*

Reasoning, multi-hop recall, and serial computation. A complementary line studies where multi-step computations occur. Geva et al. (2023) and Meng et al. (2022) localise factual recall to specific layers and modules; Yang et al. (2024) show that multi-hop reasoning is performed latently across successive layer ranges, and Biran et al. (2024) show that later hops in a chain resolve in later layers and can fail when there is insufficient depth; Brinkmann et al. (2024) and Stolfo et al. (2023) trace multi-step symbolic and arithmetic computation across layers. *These works motivate using a serial-reasoning ladder as a clean complexity axis; we measure how the internal active-head set responds to it.*

Expressivity: depth is the serial budget, heads the parallel one. The resource structure we measure is established theoretically. Merrill & Sabharwal (2023) show fixed-depth log-precision transformers lie in a shallow parallel class—unbounded fan-in but a *bounded* number of sequential steps—and Merrill & Sabharwal (2024) show that chain-of-thought decoding steps buy exactly the serial computation depth cannot supply (logarithmically many steps stay within L , polynomially many capture P); Li et al. (2024) and Feng et al. (2023) reach the same conclusion from complementary angles. Most directly, Sanford et al. (2023, 2024) prove the dichotomy we test: parallel-style problems (e.g. sparse averaging) are solvable by a single sufficiently wide layer, whereas k -hop induction and pointer chasing *require* depth growing with the hop count. *These are worst-case statements about what an architecture can compute. Our contribution is orthogonal and empirical: we measure what a trained model allocates, per task, and whether the allocation matches the task’s (D, W) structure.*

Shortcuts: models need not execute the serial algorithm. Liu et al. (2023) show transformers frequently learn shallow, parallel *shortcuts* to problems whose reference algorithm is serial, computing in far fewer sequential steps than the task nominally requires. This is the main reason a formally D -deep task may recruit no additional depth, and it makes accuracy measurement indispensable: without it, a flat depth response is ambiguous between “inside the budget”, “shortcut”

and “failed”. We incorporate this as an explicit regime of the framework (Section 3) rather than as noise.

Stages of inference and residual-stream analysis. Lad et al. (2024) argue that inference proceeds through a small number of qualitatively distinct stages (detokenisation, feature construction, prediction, sharpening), which is consistent with the conserved structure we observe (Section 5.4). The logit lens (nostalgebraist, 2020) and tuned lens (Belrose et al., 2023) read out the *content* of the residual stream by depth. *We instead study the routing: which compute units carry attribution, and how their number changes with task demand.*

Circuits and attribution. Mechanistic circuit work names specific components for specific tasks: a mathematical framework for the residual stream (Elhage et al., 2021), the indirect-object-identification circuit (Wang et al., 2022), and automated circuit discovery (Conmy et al., 2023) built on attribution patching (Nanda, 2023); Ferrando & Voita (2024) trace information-flow routes at scale. *We use the same per-head attribution criterion, but study the aggregate complement—the dead set—across a graded task family, a level of description above any single circuit.*

3 Framework: task complexity and network allocation

The framework has three parts: two coordinates describing the *task*, two describing the *network’s response*, and a law relating them.

3.1 Task side: serial depth D and parallel width W

Let G_t be the minimal dependency graph of a task t : a DAG whose nodes are the intermediate values needed to map the prompt to the answer, with an edge $u \rightarrow v$ when computing v requires u . Define

$$D(t) = \text{length of the longest directed path in } G_t \quad (\text{serial depth: the critical path}), \quad (1)$$

$$W(t) = \text{size of the largest antichain in } G_t \quad (\text{parallel width: mutually independent values}). \quad (2)$$

D counts steps that *must* happen in sequence; W counts values that must be held *simultaneously*. Both are properties of the task specification, not of any model, and in our synthetic suites both are fixed by construction (Section 4): a k -step pointer chase has $D=k, W=1$; retrieving one of m independent bindings has $D=1, W=m$.

3.2 Network side: recruited depth $\hat{D}$ and width $\hat{W}$

For a model with L layers and H heads per layer, let $s(\ell, h)$ be a per-head activity score (Section 4) and call a head *dead* when $s(\ell, h) < \tau \max_{h'} s(\ell, h')$; the complement is the *active set* $A(p) \subseteq [L] \times [H]$, and $d(\ell)$ is the per-layer dead count. Write $u = \ell/(L-1) \in [0, 1]$ for normalised depth. Define

$$\hat{W}(p) = \frac{1}{L} \sum_{\ell} (H - d(\ell)) \quad (\text{recruited width: active heads per layer}), \quad (3)$$

$$\hat{D}(p) = q_{25} \text{ of the dead-mass distribution over } u \quad (\text{recruited depth: the active frontier}). \quad (4)$$

$\hat{D}$ is a front-edge quantile—the depth by which the first quarter of dead mass has accumulated—because the *back* of the network holds a task-invariant sparse readout region that pins the centroid

and makes it blind to recruitment at the front. For cross-architecture comparison we always also report the normalised forms $\hat{W}/H$ and $\hat{D}$ (already a depth fraction); absolute head counts are not comparable between models with different H .

3.3 The resource-matching law

The hypothesis is that the model’s allocation matches the task’s structure axis-by-axis:

$$\boxed{\hat{D} = a + cD + c'W, \quad \hat{W} = b + \gamma W + \gamma'D} \quad (5)$$

with the **dissociation prediction** $c' \approx 0$ and $\gamma' \approx 0$: serial demand buys layers but not heads, parallel demand buys heads but not layers. The coefficients carry meaning beyond significance:

- $c = \text{layers per serial step}$ and $\gamma = \text{heads per parallel item}$: exchange rates converting task structure into network resource.
- $a, b =$ the **task-invariant scaffold** recruited regardless of demand (detokenisation, readout).
- $c', \gamma' = \text{leakage}$ between axes. A large γ' means the model *parallelises* a serial task (a shortcut signature); a large c' means it *serialises* a parallel task (width-to-depth conversion under capacity pressure). Both are findings, not noise.

3.4 Capacity: feasibility and the ceiling

The lattice is finite, so allocation is bounded:

$$cD \leq L_{\text{eff}}, \quad \gamma W \leq H, \quad (6)$$

where $L_{\text{eff}} = L - (\text{readout stages})$ is the usable serial budget. Equation (6) yields the framework’s sharpest prediction—a capability ceiling derived from architecture alone:

$$D_{\text{max}} \approx L_{\text{eff}} / c. \quad (7)$$

Because depth is a *bounded budget* rather than a linear resource (Merrill & Sabharwal, 2023), the expected shape of $\hat{D}(D)$ is not a line but flat-then-saturating: allocation absorbs serial steps until the budget binds. We therefore fit Eq. (5) as a *null model* and read its deviations, which—combined with accuracy—separate three regimes:

Regime	Allocation $\hat{D}$	Accuracy	Interpretation
R1 sub-ceiling ($cD \ll L_{\text{eff}}$)	rises with D (or flat if slack)	high	inside the serial budget
R2 shortcut	flat	high	model did not compute serially (Liu et al., 2023)
R3 beyond ceiling	saturated	collapses	serial budget exhausted

Accuracy is what makes R1–R3 distinguishable; a flat $\hat{D}$ alone is uninterpretable. This directly explains our Llama-3-70B result (Section 5.5): with $L=80$ and a six-step ladder, the model is deep in R1, where the framework predicts little or no depth response.

4 Method

Dead heads. For a prompt p we compute, for each attention head (ℓ, h) , a scalar attribution score $s(\ell, h)$ via gradient $\times$ activation on the head’s output (the per-head attribution of `compute_per_head_scores` in the analysis pipeline). A head is *dead* for p iff

$$s(\ell, h) < \tau \cdot \max_{h'} s(\ell, h'), \quad \tau = 0.15,$$

i.e. it carries less than 15% of the attribution of the strongest head in its own layer. The *dead-head landscape* is $d(\ell) = \#\{h : (\ell, h) \text{ dead}\}$, and the total dead count is $\sum_{\ell} d(\ell)$.

Models. Six pretrained models spanning two families and a range of depth/width: GPT-2 medium ($24\text{L} \times 16\text{H}$), large (36×20), xl (48×25); Pythia-70M (6×8), 160M (12×12), 410M (24×16).

Task ladders.

- **deep_chains** — a *pure serial-reasoning ladder*: “A is the parent of B. B is the parent of C. . . . A’s k -th descendant is” for $k = 1 \dots 6$. Each rung adds exactly one composition step; the surface template is held fixed, isolating reasoning depth.
- **complexity_gradient** — eight heterogeneous tasks ordered by nominal difficulty: lexical, subject–verb agreement, 1-hop factual, arithmetic, 2-hop, 3-hop, syllogism, analogy. Its 1/2/3-hop reasoning subset is a secondary controlled axis.

Measurements. For every (model, task) chart we report total dead heads; the dead fraction $\sum_{\ell} d(\ell)/(L \cdot H)$; the dead-mass *centroid* in fractional depth (0 = first layer, 1 = last); the fraction of dead mass in the front half of the network; and the *front-edge depth* q_{25} —the normalised depth at which the first quarter of dead mass accumulates. q_{25} is our primary horizontal-axis statistic because, unlike the centroid, it is not dominated by the task-invariant sparse readout region at the back of the network and so is sensitive to where the *active frontier* sits. The complete set of $6 \times 8 + 6 \times 6 = 84$ measurements is listed in Appendix A.

Auxiliary exact data. To corroborate the figure-recovered trends with ground-truth attribution we additionally analyse one model, Llama-3-8B ($32\text{L} \times 32\text{H}$), for which exact per-head dead masks are available from the analysis pipeline (read directly from the per-head graph files, not recovered from a rendered chart). We use it only as a high-fidelity check on the two-axis correlations.

Data provenance and recovery. The sweep was archived as rendered per-layer bar charts (the `abdulla_results/` directory; subfolders `graphs`, `graphs 2`, `graphs 3`). Because the underlying numeric arrays were not saved, we recover per-layer counts directly from the images with a calibrated extractor (`extract_dead_heads.py`): the y -axis is calibrated from the rendered “total heads = N ” line and the plot baseline, and each layer’s bar height is read as a contiguous run anchored at the baseline. Recovered values were validated against the printed bar labels on GPT-2 xl and GPT-2 large to within ± 1 –2 heads. We treat the recovered data as reliable for trend analysis; regenerating exact arrays from the pipeline is the first item on the roadmap (Section 9).

5 Results

5.1 Complexity-graded head recruitment

Table 1 reports total dead heads on the serial-reasoning ladder. Across the larger models the count falls as reasoning depth rises: the model engages more heads to perform more composition steps. The effect is strongest and most monotone in the deepest/widest models (GPT-2 xl decreases on four of five rungs, a net -131 heads from 1-hop to 6-hop; Pythia-410M a net -24).

Table 1: Total dead attention heads on the `deep_chains` serial-reasoning ladder. Fewer dead heads = more heads recruited into active computation.

Model	1-hop	2-hop	3-hop	4-hop	5-hop	6-hop
GPT-2 medium	29	15	11	9	15	15
GPT-2 large	70	34	45	38	47	48
GPT-2 xl	228	142	115	105	113	97
Pythia-70M	3	2	1	1	4	2
Pythia-160M	22	10	11	16	23	18
Pythia-410M	66	53	56	47	46	42

A universal first-hop cliff. The single largest recruitment event is the 1-hop $\rightarrow$ 2-hop step—the introduction of the first genuine composition. It recruits a large fraction of all idle heads in *every* model:

Model	1 $\rightarrow$ 2 recruited	Model	1 $\rightarrow$ 2 recruited
GPT-2 medium	+48%	Pythia-70M	+33%
GPT-2 large	+51%	Pythia-160M	+55%
GPT-2 xl	+38%	Pythia-410M	+20%

The same ordering appears in the heterogeneous complexity gradient (Table 2): the surface/lexical, single-hop-factual, and analogy tasks leave the most heads dead, while the multi-step 2-hop and 3-hop reasoning tasks engage the most heads (e.g. GPT-2 xl: lexical 311 vs. 2-hop 142, 3-hop 115).

Table 2: Total dead heads on the `complexity_gradient` suite. Easy surface tasks leave the most heads dead; multi-step reasoning engages the most.

Model	Lexical	Subj-verb	1-hop fact	Arithmetic	2-hop	3-hop	Syllogism	Analogy
GPT-2 medium	60	37	57	38	15	11	35	38
GPT-2 large	106	71	113	94	34	45	101	120
GPT-2 xl	311	229	252	212	142	115	228	289
Pythia-70M	7	4	4	4	2	1	3	2
Pythia-160M	31	20	22	21	10	11	17	14
Pythia-410M	69	71	64	45	53	56	69	75

5.2 The recruitable reserve dilates with scale

Table 3 expresses the same data as a fraction of all heads. Two scale effects are visible. First, larger models carry a larger idle reserve on the easiest task (GPT-2: 8% $\rightarrow$ 10% $\rightarrow$ 19% for medium $\rightarrow$ large $\rightarrow$ xl at 1-hop). Second, they exercise a wider dynamic range across the ladder (drawing down 3.7, 3.1, and 10.9 percentage points respectively). The model behaves as though

capacity is a reserve held against demand: the more capacity, the more is held idle when it is not needed, and the more can be mobilised when it is.

Table 3: Dead heads as a fraction of all heads on **deep_chains**.

Model	1-hop	2-hop	3-hop	4-hop	5-hop	6-hop
GPT-2 medium	8%	4%	3%	2%	4%	4%
GPT-2 large	10%	5%	6%	5%	7%	7%
GPT-2 xl	19%	12%	10%	9%	9%	8%
Pythia-70M	6%	4%	2%	2%	8%	4%
Pythia-160M	15%	7%	8%	11%	16%	12%
Pythia-410M	17%	14%	14%	12%	12%	11%

5.3 The horizontal axis: the active frontier extends with demand

Recruitment has not only a magnitude but a *direction*. We ask where in depth the recruited heads come from by tracking the front edge of the dead-head distribution—the normalised depth q_{25} at which the first quarter of dead mass has accumulated (a front-sensitive statistic, unlike the centroid, which is anchored by the task-invariant sparse readout region at the back). As reasoning depth rises, q_{25} moves deeper: the dead zone retreats and the *active frontier* advances. Table 4 reports both axes together as correlations with hop count.

Table 4: The two coupled axes on **deep_chains**, plus the exact Llama-3-8B reasoning subset (1/2/3-hop, per-head attribution, not figure-recovered). Negative $r(\text{hops}, \text{total})$ is recruitment (vertical axis); positive $r(\text{hops}, q_{25})$ and $r(\text{hops}, \text{centroid})$ is frontier extension (horizontal axis). Both effects appear in the deep/wide models and strengthen with scale; the smallest models saturate.

Model	Geom (L×H)	$r(\text{hops}, \text{total})$	$r(\text{hops}, q_{25})$	$r(\text{hops}, \text{centroid})$
GPT-2 xl	48×25	−0.82	+0.48	+0.78
GPT-2 large	36×20	−0.33	+0.88	+0.29
Pythia-410M	24×16	−0.93	+0.72	+0.66
GPT-2 medium	24×16	−0.55	−0.34	−0.32
Pythia-160M	12×12	+0.24	−0.61	−0.59
Pythia-70M	6×8	+0.05	−0.65	−0.65
Llama-3-8B	32×32	−0.89	+0.96	+1.00

The signal is clearest in the model with exact (non-recovered) attribution. On Llama-3-8B the three-rung reasoning ladder shows total dead falling ($425 \rightarrow 256 \rightarrow 248$), early-layer ($\ell \leq 10$) dead falling sharply ($92 \rightarrow 47 \rightarrow 42$), and the frontier advancing monotonically ($q_{25} : 0.46 \rightarrow 0.54 \rightarrow 0.58$; centroid $0.67 \rightarrow 0.71 \rightarrow 0.76$). Among the figure-recovered models the same coupling holds for the three largest (GPT-2 xl/large, Pythia-410M): $r(\text{hops}, q_{25})$ pools to +0.70 ($n=18$). The three smallest models do not show it—their counts are quantisation-noisy and, with only 6–24 layers, they appear to exhaust their reserve within one or two hops, so neither axis has room to move. This is the same scale dependence seen for recruitment (Section 5.2): both axes require spare capacity to be visible.

A per-hop depth cost. Converting the frontier shift to absolute layers gives a serial cost per reasoning step: ≈ 1.9 layers/hop on Llama-3-8B (1→3 hop), and 0.3–0.5 layers/hop on the larger GPT-2/Pythia chains. The smaller per-hop cost on the GPT-2 chains is consistent with early

saturation—the frontier stops advancing once the model can no longer carry out additional hops—which predicts that the frontier should track task *accuracy*, a test we make explicit in the roadmap (Section 9). This per-hop layer cost is the empirical, per-layer counterpart of the theoretical result that transformer depth bounds serial computation (Merrill & Sabharwal, 2023; Li et al., 2024) and the finding that later hops resolve in later layers and fail without sufficient depth (Yang et al., 2024; Biran et al., 2024).

5.4 A conserved backbone

The recruitment that does occur is a deformation of a stable structure, not a redrawing of it. Most dead mass sits in the back half of the network at every difficulty level (front-half fractions are small and roughly constant; see Appendix A): the late layers keep attention sparse regardless of task, while recruitment is concentrated in the early-to-middle band. This conserved-then-deformed picture is consistent with the view that inference proceeds through a small set of qualitatively distinct stages (Lad et al., 2024).

5.5 Controlled dissociation: a (D, W) factorial

The complexity gradient and deep chains confound *what* kind of demand rises with *how much*. To test the sharper hypothesis—that *serial* demand (dependency depth D) recruits network *depth* while *parallel* demand (independent width W) recruits attention *heads* (Figure 1)—we built a factorial task suite that varies D and W independently: a pure-serial ladder (a constant-length pointer-chase, $D=1 \dots 6$, $W=1$), a pure-parallel ladder (find a unique binding among W items, $D=1$, $W=1 \dots 6$, with a length-matched variant), and a crossed 3×3 grid. We measure, per task, the recruited *width* $\hat{W}$ (active heads per layer) and recruited *depth* $\hat{D}$ (the active-frontier depth, i.e. the layer quantiles of the dead-mass distribution). Results are on Llama-3-70B¹ and should be read as a proof-of-method: the effects are small because, as Section 5.2 shows, a 70B holds an enormous idle reserve that compresses all recruitment.

Width recruits heads; depth does not. On the length-matched parallel ladder, active heads per layer rise with W ($r=+0.81$; Figure 3 left), while the active frontier does not move ($r(W, q_{25})=-0.09$). The crossed grid confirms and separates the two axes: fitting active-heads-per-layer gives $\approx 32.3 - 0.14D + 0.25W$ —a clean positive width coefficient and a near-zero depth coefficient (Figure 2)—while the depth coefficient on the frontier is ≈ 0.0005 (in depth fraction), i.e. zero. **Parallel width engages more heads at fixed depth; serial depth does not engage more heads.** This is the width half of the predicted double dissociation.

The length control is necessary. The *raw* parallel ladder shows active heads *falling* with W —the opposite of the prediction—but this reverses once prompt length is held constant (Figure 3 left, dashed vs. solid). The raw trend is a length artifact (a longer prompt leaves more heads below threshold at the final token); only the length-matched ladder reveals the true width effect. We therefore treat length matching as mandatory for the width axis.

The depth axis is not resolved on the 70B (a null, not a refutation). Serial depth produces *no* detectable frontier extension here: across $D=1 \dots 6$ the front-edge, centroid, and back-edge of the active region each shift by less than one layer on an 80-layer network (Figure 3 right)—noise.

¹Meta-Llama-3-70B, loaded in bfloat16 with CPU offload; forward-only magnitude metric.

Hypothesis: serial demand $\rightarrow$ depth, parallel demand $\rightarrow$ width

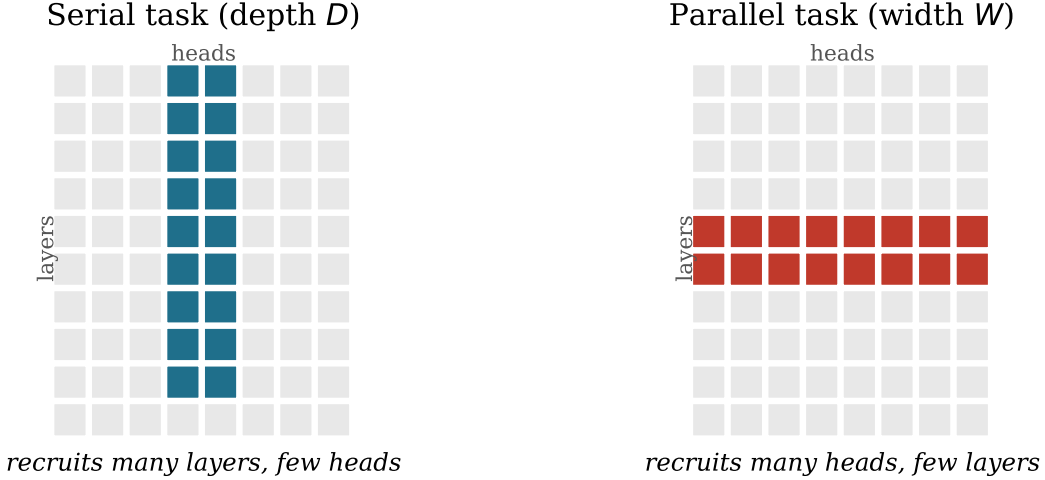


Figure 1: The hypothesis. A serial task should recruit a deep, narrow band of units (many layers, few heads); a parallel task a shallow, wide band (many heads, few layers). The active subgraph is the task’s computation graph embedded into the layer $\times$ head lattice.

We read this as an instrument problem, not evidence against the hypothesis. A 70B idles $\sim$ half its heads (Section 5.2), so a six-step task is trivially within capacity and barely perturbs the geometry; and without accuracy measurements we cannot verify the model computes the pointer-chase serially at all. The depth axis must be tested where the reserve is small—the GPT-2/Pythia models, on which six hops is a large fraction of total depth—with per-cell accuracy and paraphrase replicates. We state this explicitly so the confirmed width axis is not over-read into a full dissociation.

6 Architecture sets the two budgets

Equations (6)–(7) make the framework architecture-sensitive: the serial budget is set by depth, the parallel budget by the head structure, and their ratio by the model’s *aspect ratio* L/H . Table 5 lists the relevant geometry for the model families in and around this study.

Predictions. The framework converts these differences into falsifiable statements. We state them as pre-registered predictions; the runs that test them are itemised in Section 9.

- A1. The parallel budget is head-structured.** γ (heads per parallel item) should scale with the number of *independent* attention channels. Under multi-head attention that is H ; under grouped-query attention, where many query heads share one KV group, the effective budget may be closer to n_{kv} . Prediction: width recruitment saturates near n_{kv} , not H , in GQA models.
- A2. The serial budget is depth.** Either c is roughly constant in *absolute layers* across models (so deeper models simply fit more steps), or constant as a *fraction* of L (so all models exhaust their budget at the same D). These give very different $D_{\max}$ predictions via Eq. (7), and discriminating them is the central cross-model measurement.

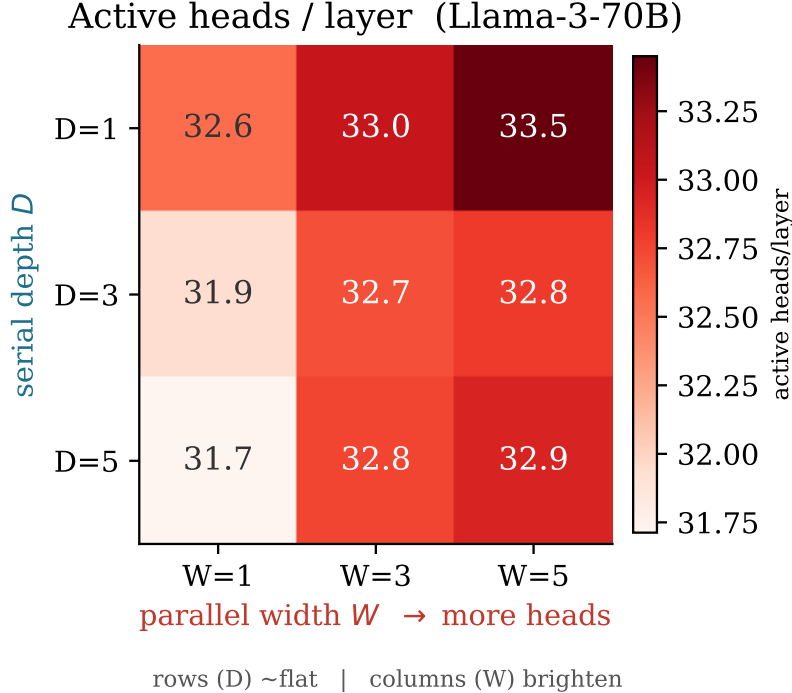


Figure 2: Active heads per layer across the crossed (D, W) grid on Llama-3-70B. Columns brighten with parallel width W (more heads engaged); rows are flat across serial depth D . Fit: $32.3 - 0.14D + 0.25W$.

- A3. Block topology should not matter for the depth axis.** Pythia computes attention and MLP *in parallel* within a block, GPT-2 sequentially. At matched (L, H) —Pythia-410M vs. GPT-2-medium, both 24×16 —the framework predicts the same c , i.e. the depth axis is a property of layer count rather than intra-block wiring.
- A4. Aspect ratio beats parameter count for serial tasks.** Pythia-1B (16×8) is *shallower and narrower* than Pythia-410M (24×16) despite having more parameters. Prediction: 1B hits its serial ceiling *earlier* than 410M—a within-family dissociation of capability from scale.
- A5. The width axis is attention-specific; the depth axis is not.** State-space models have layers but no heads. Prediction: the depth axis survives in Mamba-2/RWKV (serial demand still recruits layers) while the width axis has no attentional analogue—separating the two axes by architecture rather than by task. Correspondingly, in mixture-of-experts models the width axis should extend into *expert* space (parallel demand recruits more experts per layer).

What the present data can and cannot say. Our measurements span two of these cells and are suggestive rather than decisive. (i) Both GPT-2 (sequential) and Pythia (parallel-block) families show demand-graded recruitment and, in their larger members, frontier extension (Sections 5.1, 5.3), consistent with A3 but not a matched- (L, H) test. (ii) The Llama-3 models are the only GQA systems here, and their unusually large idle fraction ($\sim 50\%$ of *query* heads inactive on the 70B under forward-only metrics) is consistent with an effective parallel budget nearer $n_{kv}=8$ than $H=64$ —but this is a single-architecture observation, confounded with scale, and we flag it as a hypothesis for A1 rather than evidence for it. (iii) No SSM or MoE model was measured. A1–A5 are therefore

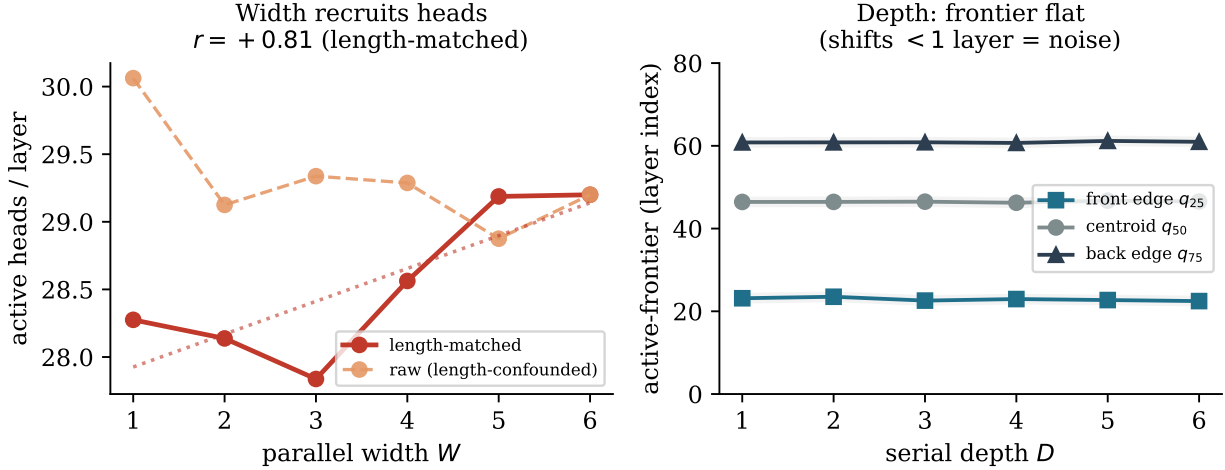


Figure 3: The two axes on Llama-3-70B. **Left:** parallel width recruits heads (active heads/layer vs. W , length-matched, $r=+0.81$); the raw (non-length-matched) curve trends the opposite way—a length artifact that the control removes. **Right:** serial depth does *not* move the active frontier: the front/centroid/back quantiles shift by < 1 layer across $D=1 \dots 6$ (grey band = ± 1 layer), i.e. within noise.

Table 5: Architectural determinants of the two budgets. n_{kv} is the number of key/value groups (equal to H under multi-head attention, smaller under grouped-query attention). “Block” distinguishes sequential attention→MLP from parallel attention+MLP.

Family	Model	L	H	n_{kv}	Block / notes
GPT-2	small / medium / large / xl	12–48	12–25	$= H$	sequential, MHA
Pythia	70M / 160M / 410M	6–24	8–16	$= H$	parallel attn+MLP
Pythia	1B	16	8	$= H$	aspect-ratio inversion (see A4)
Pythia	1.4B / 2.8B	24 / 32	16 / 32	$= H$	parallel attn+MLP
Llama-3	8B	32	32	8	sequential, GQA
Llama-3	70B	80	64	8	sequential, GQA
Mixtral / OLMoE	MoE	—	—	—	experts: a second width axis
Mamba-2 / RWKV	SSM	—	<i>none</i>	—	no heads: width axis undefined

predictions of the framework, not results of this paper.

7 Discussion

The measurements support a compact two-axis account. Treat each model as holding a reserve of dormant attention heads that it *recruits* on demand.

Vertical axis (recruitment). Rising task demand—most cleanly, additional reasoning steps—is paid for by activating more of this reserve, so $d(\ell)$ deflates monotonically. The size of the reserve, and how much of it is mobilised, are set by model capacity: larger models hold more idle width and mobilise more of it under load.

Horizontal axis (frontier extension). The recruitment is not spatially uniform. Each additional composition step pulls the active frontier deeper into the network: early and middle layers that idle on a shallow task are engaged on a deeper one, and the front edge of the dead zone retreats

(Section 5.3). This advance has an estimable per-hop layer cost and appears to plateau when the model runs out of usable depth, linking the geometry directly to the serial-depth limits of transformers (Merrill & Sabharwal, 2023; Yang et al., 2024; Biran et al., 2024).

The two axes are coupled—more demand both deflates the landscape and advances its frontier—and both are gated by spare capacity, which is why they emerge in the larger models and vanish into noise in the smallest. The deformation acts on a conserved backbone: a task-invariant sparse readout region at the back persists at every difficulty, while the action happens in the early-to-middle band. In short: a capacity-gated recruitment response with a magnitude (how many heads) and a reach (how deep).

7.1 What active-node geometry says about circuit structure

A circuit is a set of *paths*—head reading from head, composing into a readout. Our observable is a set of *nodes*. The relationship is a marginal, and being precise about it determines what may be claimed.

What node activity licenses. Under the faithfulness assumption that dead units are dispensable (the ablation check in Section 9, item D’), the active set *contains* the circuit’s nodes. Node activity therefore constrains: (i) the circuit’s **support**—an over-approximation of where it lives; (ii) its **geometry**—width per layer, depth-extent, total size; and (iii) its **schedule**—which layers are working at all. In the path-bundle picture, $\hat{W}$ is the bundle’s layer-wise width and $\hat{D}$ is how deep the bundle reaches, which is exactly why (D, W) maps onto $(\hat{D}, \hat{W})$: serial structure lengthens the bundle, parallel structure widens it.

What it does not license. Node activity is silent about **wiring**. Two circuits with different connectivity can activate identical nodes, so we cannot say from $d(\ell)$ which active head reads from which earlier head. Edge-level methods are required for that: path patching (Goldowsky-Dill et al., 2023), automated circuit discovery (Conmy et al., 2023; Syed et al., 2023), and information-flow routes (Ferrando & Voita, 2024), of which our node counts are a cheap marginal. We therefore make no wiring claims.

The task prior is what closes the gap. Node geometry alone underdetermines a circuit—but we never have it alone. We have it *paired with a task whose computational structure we designed*. Nodes plus a known G_t identify the **embedding** of G_t into the layer×head lattice: if a k -hop task is known to require k sequential steps and we observe k successive active bands, the task prior licenses the inference that each band carries one hop—an inferred *schedule*, even though no edge was measured. This is the precise sense in which task structure is readable from activation structure, and it is weaker than circuit discovery by design.

A two-way correspondence. The framework is thus a correspondence between task structure and active-region geometry, usable in both directions. *Forward*: (D, W) predicts $(\hat{D}, \hat{W})$ —the law of Eq. (5). *Inverse*: the geometry of the active region decodes the task’s computational type, making the active subgraph a **fingerprint of the computation**. The inverse direction is attractive precisely because it requires no wiring claim: showing that a simple decoder recovers (D, W) from activation geometry demonstrates that node activity *carries* structural information about the computation, whatever the wiring turns out to be.

Relation to circuit discovery. The two approaches are complementary rather than competing. Circuit discovery recovers wiring for *one* task at high cost; we recover geometry for *many* tasks cheaply, and—because the task family is graded—obtain a scaling law that single-circuit studies cannot express. Most mechanistic work asks “what is the circuit for task X ”; we ask “how does circuit *shape* vary with the structure of X ”. Studies of circuit reuse (Merullo et al., 2024), function vectors (Todd et al., 2024), and specialised head families—induction (Olsson et al., 2022), retrieval (Wu et al., 2024), successor (Gould et al., 2023)—supply the component vocabulary that a wiring-level version of this account would use.

7.2 Implications

Conditional computation has a second knob. Adaptive-depth and early-exit methods (Graves, 2016; Elbayad et al., 2020; Schuster et al., 2022; Raposo et al., 2024) adapt one scalar: how deep to go. If parallel demand is carried by heads and serial demand by layers, then *width* is an equally principled axis for conditional computation, and γ estimates how much width a given parallel load requires. A demand-aware model would taper both.

Capability is predictable from shape, not only size. Eq. (7) says the maximum serial depth a model can execute in one forward pass is governed by L_{eff}/c . Two models with equal parameter counts but different aspect ratios should therefore differ systematically in serial-task competence (prediction A4)—capability tied to architectural *shape* rather than scale.

Where chain-of-thought enters. If serial depth beyond the in-context budget must be externalised, the natural continuation of the depth axis is across *tokens*: past D_{max} , serial computation should migrate from layers to chain-of-thought steps, each running a bounded-depth circuit (Merrill & Sabharwal, 2024; Li et al., 2024). Measuring that migration—serial structure moving from the depth axis to the token axis at the ceiling—is the most direct extension of this work and we leave it to follow-up.

8 Limitations

1. **Data recovered from figures.** Counts are recovered from rendered bar charts (Section 4); they are validated to ± 1 –2 heads but are not the exact arrays. Regenerating numeric data from the pipeline is the first roadmap item.
2. **Small models are noisy.** Pythia-70M/160M have few heads, so their counts are quantisation-noisy and some rungs are non-monotone (Table 1); the clean trends rest on the larger models.
3. **Single threshold.** “Dead” is defined at $\tau = 0.15$; the robustness of the trend to τ must be demonstrated.
4. **Width and reasoning depth are not separated.** The `deep_chains` ladder raises reasoning depth and, incidentally, the number of intermediate entities to hold; a width-only control is required to attribute the vertical axis to demand and the horizontal axis to serial depth specifically.
5. **The horizontal axis rests partly on recovered data.** The frontier correlations are strongest and cleanest on the exact Llama-3-8B data ($r=+0.96$) but modest on individual figure-recovered models; the pooled estimate ($r=+0.70$) and the exact check are what carry it. It should be reproduced on exact data across the full sweep.

6. **No behavioural correlate yet.** We have not tied either axis to task accuracy; the prediction is that both recruitment and frontier advance track whether the model can actually perform the task, and that the frontier plateau coincides with the accuracy collapse.
7. **Single prompt per condition.** No paraphrase variance is reported, so surface-length confounds are not yet excluded. With one prompt per (D, W) cell, the coefficients of Eq. (5) are point estimates without intervals and should not be quoted as measured exchange rates.
8. **Node activity is a marginal of the circuit.** Our observable constrains a circuit’s support and geometry but not its wiring (Section 7.1); all structural claims are therefore about *shape*, not connectivity.
9. **Heads may not be the right parallel unit.** Attention heads are polysemantic and features are superposed, so “one parallel item $\approx \gamma$ heads” may be diffuse rather than crisp; a feature-level version of $\hat{W}$ could behave differently.
10. **Cross-architecture claims are predictions, not results.** A1–A5 (Section 6) follow from the framework but are untested here: no MoE or state-space model was measured, and the GQA observation is confounded with scale.
11. **Correlational, not causal.** c and γ describe allocation; only ablation shows the recruited units are *necessary* (roadmap item D’).

9 Roadmap: experiments to finish the paper

The following is an ordered, concrete plan to convert these observations into a strong submission.

A. Regenerate exact numeric data (removes Limitation 1). Add a `-dump-dead-matrix` option to the analysis pipeline that writes the full per- (ℓ, h) attribution and the binary dead mask to CSV/JSON for every (model, task), and rerun the six-model sweep. All tables in this paper should then be regenerated from exact arrays.

B. Dissociate recruitment from reasoning depth (the key control). Add a *width-control* suite that raises parallel demand at fixed reasoning depth—e.g. retrieve k *independent*, non-chained facts in one prompt for $k = 1 \dots 6$. Prediction: dead-head count falls with k (more heads recruited) even though no additional composition step is added. This is the experiment that establishes recruitment as a demand response rather than a depth artefact.

C. Tie both axes to behaviour. Score every model on every rung of every ladder (exact-match accuracy on the final entity). Test two predictions: (i) recruitment co-varies with success—where a model cannot perform the task, recruitment stalls; and (ii) the *active frontier plateaus at the depth where accuracy collapses*, making the frontier a read-out of the model’s usable serial depth. Report, per model, the correlation of accuracy with both dead-head count and frontier depth q_{25} .

C’. Confirm the horizontal axis on exact data across the sweep. The frontier effect is clean on exact Llama-3-8B attribution but rests on pooled/recovered estimates for the GPT-2/Pythia models. Recompute q_{25} , the centroid, and early-layer dead counts from the exact masks of item A for all models and rungs, and report per-model $r(\text{hops}, q_{25})$ with confidence intervals. This converts the headline two-axis result from “suggestive across models, exact on one” to “exact across all.”

D'. Faithfulness: validate that “active” means “necessary”. On models small enough for causal analysis (GPT-2 small/medium with TransformerLens), ablate the heads we call dead and verify behaviour is preserved, then ablate the active set and verify it breaks—the faithfulness–sparsity curve of Conmy et al. (2023), using activation or attribution patching (Nanda, 2023; Syed et al., 2023) against a proper corrupted baseline and reporting faithfulness *and* completeness (Hanna et al., 2024; Makelov et al., 2024). This single experiment does three jobs: it makes the active set causal rather than correlational, it *defines* τ operationally (the elbow of the curve), and it licenses the support claim of Section 7.1. It is a prerequisite for the structural language used throughout.

D. Threshold robustness. Recompute all results for $\tau \in \{0.05, 0.10, 0.15, 0.20\}$ and show the recruitment trend (sign and approximate magnitude) is invariant. Include as a supplementary figure.

E. Paraphrase and statistical rigour. Generate ≥ 5 surface paraphrases per condition (re-named entities, reordered clauses, matched token length) and report means with confidence intervals and a per-model Spearman $\rho(\text{demand}, \text{dead-count})$. This rules out length/surface confounds and quantifies significance.

F. Extend to instruction-tuned and larger models. Repeat the cleanest ladder (`deep_chains`, `width-control`) on at least one modern model family (e.g. a Llama-3 or Pythia-1.4B/2.8B series) to test whether the scale-dilation trend continues; restoring per-head attribution on the large-model path is a prerequisite.

G. Head-identity analysis (optional, strengthens mechanism). Track *which* heads are recruited as demand rises—are they consistent across prompts, do early-recruited heads correspond to known functional types? This connects the aggregate count to the circuits literature.

H. Two headline figures. (i) A small-multiples panel of $d(\ell)$ for all six models $\times$ six rungs, with the conserved sparse readout region shaded, so the recruitment-and-scale story is legible at a glance. (ii) A two-axis summary: per model, total dead (vertical) and frontier depth q_{25} (horizontal) as functions of hop count, so the coupled deformation and its scale dependence are visible together.

Deliverables checklist. exact-data regeneration (A) $\rightarrow$ width control (B) $\rightarrow$ behaviour coupling (C) $\rightarrow$ exact two-axis confirmation (C') $\rightarrow$ robustness + statistics (D, E) $\rightarrow$ scale extension (F) $\rightarrow$ optional mechanism (G) $\rightarrow$ headline figures (H). Items A–E and C' are necessary for a credible submission; F–H strengthen it.

10 Conclusion

We introduced the dead-head landscape as a graded probe of how transformers allocate their attention machinery to a task, and established a two-axis demand–geometry law. As task demand rises the landscape deforms in two coupled ways: it *deflates*—models monotonically engage dormant heads, drawing on a reserve whose size scales with capacity (vertical recruitment)—and its *active frontier advances*—each additional reasoning step pulls computation deeper into the network at an estimable per-hop layer cost (horizontal extension). Both effects are gated by spare capacity, emerging in larger models and washing out in the smallest. The full dataset and a concrete experimental roadmap are provided so the result can be completed and validated.

A Complete dataset

All 84 measurements recovered from `abdulla_results/`. “Frac” is dead heads over total heads; “ q_{25} ” is the front-edge depth (horizontal axis: normalised depth at which the first quarter of dead mass accumulates); “Centroid” is the dead-mass centre in fractional depth (0 = first layer, 1 = last); “Front” is the fraction of dead mass in the first half of the network.

A.1 complexity_gradient suite (48 measurements)

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
GPT-2 medium	24×16	Lexical	60	15.5%	0.69	0.78	0.05
GPT-2 medium	24×16	Subj-verb	37	9.6%	0.84	0.88	0.03
GPT-2 medium	24×16	1-hop fact	57	14.7%	0.48	0.71	0.21
GPT-2 medium	24×16	Arithmetic	38	9.8%	0.58	0.76	0.08
GPT-2 medium	24×16	2-hop	15	3.9%	0.70	0.82	0.07
GPT-2 medium	24×16	3-hop	11	2.8%	0.58	0.80	0.09
GPT-2 medium	24×16	Syllogism	35	9.0%	0.71	0.80	0.06
GPT-2 medium	24×16	Analogy	38	9.8%	0.59	0.74	0.08
GPT-2 large	36×20	Lexical	106	14.7%	0.67	0.76	0.02
GPT-2 large	36×20	Subj-verb	71	9.9%	0.70	0.81	0.00
GPT-2 large	36×20	1-hop fact	113	15.7%	0.49	0.69	0.25
GPT-2 large	36×20	Arithmetic	94	13.1%	0.62	0.74	0.12
GPT-2 large	36×20	2-hop	34	4.8%	0.62	0.75	0.08
GPT-2 large	36×20	3-hop	45	6.3%	0.66	0.78	0.06
GPT-2 large	36×20	Syllogism	101	14.1%	0.72	0.78	0.03
GPT-2 large	36×20	Analogy	120	16.7%	0.63	0.74	0.04
GPT-2 xl	48×25	Lexical	311	25.9%	0.64	0.74	0.05
GPT-2 xl	48×25	Subj-verb	229	19.1%	0.67	0.75	0.03
GPT-2 xl	48×25	1-hop fact	252	21.0%	0.54	0.66	0.18
GPT-2 xl	48×25	Arithmetic	212	17.7%	0.55	0.66	0.16
GPT-2 xl	48×25	2-hop	142	11.9%	0.62	0.71	0.07
GPT-2 xl	48×25	3-hop	115	9.6%	0.62	0.73	0.07
GPT-2 xl	48×25	Syllogism	228	19.0%	0.65	0.74	0.04
GPT-2 xl	48×25	Analogy	289	24.1%	0.62	0.72	0.09
Pythia-70M	6×8	Lexical	7	14.4%	0.64	0.81	0.00
Pythia-70M	6×8	Subj-verb	4	8.3%	0.74	0.86	0.00
Pythia-70M	6×8	1-hop fact	4	8.3%	0.74	0.86	0.00
Pythia-70M	6×8	Arithmetic	4	8.3%	0.74	0.86	0.00
Pythia-70M	6×8	2-hop	2	4.1%	0.90	0.90	0.00
Pythia-70M	6×8	3-hop	1	2.0%	0.90	0.90	0.00
Pythia-70M	6×8	Syllogism	3	6.2%	0.90	0.90	0.00
Pythia-70M	6×8	Analogy	2	4.1%	0.90	0.90	0.00
Pythia-160M	12×12	Lexical	31	21.3%	0.62	0.70	0.13
Pythia-160M	12×12	Subj-verb	20	13.7%	0.72	0.81	0.00
Pythia-160M	12×12	1-hop fact	22	15.1%	0.60	0.75	0.09
Pythia-160M	12×12	Arithmetic	21	14.4%	0.64	0.81	0.00
Pythia-160M	12×12	2-hop	10	6.8%	0.64	0.75	0.00
Pythia-160M	12×12	3-hop	11	7.5%	0.65	0.76	0.00
Pythia-160M	12×12	Syllogism	17	11.7%	0.58	0.75	0.00

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
Pythia-160M	12×12	Analogy	14	9.6%	0.65	0.75	0.00
Pythia-410M	24×16	Lexical	69	18.1%	0.67	0.77	0.04
Pythia-410M	24×16	Subj-verb	71	18.6%	0.69	0.79	0.03
Pythia-410M	24×16	1-hop fact	64	16.8%	0.65	0.74	0.12
Pythia-410M	24×16	Arithmetic	45	11.6%	0.79	0.84	0.02
Pythia-410M	24×16	2-hop	53	13.9%	0.62	0.74	0.15
Pythia-410M	24×16	3-hop	56	14.5%	0.59	0.73	0.16
Pythia-410M	24×16	Syllogism	69	18.1%	0.71	0.78	0.10
Pythia-410M	24×16	Analogy	75	19.6%	0.64	0.74	0.07

A.2 deep_chains suite (36 measurements)

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
GPT-2 medium	24×16	1-hop chain	29	7.5%	0.64	0.78	0.07
GPT-2 medium	24×16	2-hop chain	15	3.9%	0.70	0.82	0.07
GPT-2 medium	24×16	3-hop chain	11	2.8%	0.58	0.80	0.09
GPT-2 medium	24×16	4-hop chain	9	2.3%	0.38	0.66	0.44
GPT-2 medium	24×16	5-hop chain	15	3.8%	0.61	0.76	0.13
GPT-2 medium	24×16	6-hop chain	15	3.8%	0.60	0.74	0.20
GPT-2 large	36×20	1-hop chain	70	9.8%	0.63	0.74	0.04
GPT-2 large	36×20	2-hop chain	34	4.8%	0.62	0.75	0.08
GPT-2 large	36×20	3-hop chain	45	6.3%	0.66	0.78	0.06
GPT-2 large	36×20	4-hop chain	38	5.3%	0.65	0.78	0.10
GPT-2 large	36×20	5-hop chain	47	6.6%	0.67	0.78	0.04
GPT-2 large	36×20	6-hop chain	48	6.7%	0.68	0.78	0.02
GPT-2 xl	48×25	1-hop chain	228	19.0%	0.62	0.72	0.08
GPT-2 xl	48×25	2-hop chain	142	11.9%	0.62	0.71	0.07
GPT-2 xl	48×25	3-hop chain	115	9.6%	0.62	0.73	0.07
GPT-2 xl	48×25	4-hop chain	105	8.7%	0.61	0.72	0.08
GPT-2 xl	48×25	5-hop chain	113	9.4%	0.62	0.74	0.03
GPT-2 xl	48×25	6-hop chain	97	8.1%	0.65	0.74	0.02
Pythia-70M	6×8	1-hop chain	3	6.2%	0.90	0.90	0.00
Pythia-70M	6×8	2-hop chain	2	4.1%	0.90	0.90	0.00
Pythia-70M	6×8	3-hop chain	1	2.0%	0.90	0.90	0.00
Pythia-70M	6×8	4-hop chain	1	2.0%	0.90	0.90	0.00
Pythia-70M	6×8	5-hop chain	4	8.3%	0.90	0.90	0.00
Pythia-70M	6×8	6-hop chain	2	4.1%	0.74	0.82	0.00
Pythia-160M	12×12	1-hop chain	22	15.1%	0.61	0.74	0.04
Pythia-160M	12×12	2-hop chain	10	6.8%	0.64	0.75	0.00
Pythia-160M	12×12	3-hop chain	11	7.5%	0.65	0.76	0.00
Pythia-160M	12×12	4-hop chain	16	11.0%	0.60	0.76	0.00
Pythia-160M	12×12	5-hop chain	23	15.8%	0.60	0.73	0.00
Pythia-160M	12×12	6-hop chain	18	12.4%	0.58	0.73	0.00
Pythia-410M	24×16	1-hop chain	66	17.3%	0.58	0.73	0.12
Pythia-410M	24×16	2-hop chain	53	13.9%	0.62	0.74	0.15
Pythia-410M	24×16	3-hop chain	56	14.5%	0.59	0.73	0.16
Pythia-410M	24×16	4-hop chain	47	12.4%	0.57	0.71	0.12
Pythia-410M	24×16	5-hop chain	46	11.9%	0.67	0.77	0.11

Model	Geom (L×H)	Task	Dead	Frac	q_{25}	Centroid	Front
Pythia-410M	24×16	6-hop chain	42	10.8%	0.68	0.78	0.12

References

- N. Belrose et al. Eliciting latent predictions from transformers with the tuned lens. *arXiv:2303.08112*, 2023.
- J. Brinkmann et al. A mechanistic analysis of a transformer trained on a symbolic multi-step reasoning task. *ACL*, 2024.
- N. Goldowsky-Dill et al. Localizing model behavior with path patching. *arXiv:2304.05969*, 2023.
- R. Gould et al. Successor heads: Recurring, interpretable attention heads in the wild. *ICLR*, 2024 (arXiv 2023).
- M. Hanna, S. Pezzelle, Y. Belinkov. Have faith in faithfulness: Going beyond circuit overlap when finding model mechanisms. *arXiv:2403.17806*, 2024.
- B. Liu et al. Transformers learn shortcuts to automata. *ICLR*, 2023.
- A. Makelov, G. Lange, N. Nanda. Is this the subspace you are looking for? An interpretability illusion for subspace activation patching. *ICLR*, 2024.
- W. Merrill, A. Sabharwal. The expressive power of transformers with chain of thought. *ICLR*, 2024.
- J. Merullo, C. Eickhoff, E. Pavlick. Circuit component reuse across tasks in transformer language models. *ICLR*, 2024.
- C. Olsson et al. In-context learning and induction heads. *Anthropic*, 2022.
- C. Sanford, D. Hsu, M. Telgarsky. Representational strengths and limitations of transformers. *NeurIPS*, 2023.
- C. Sanford, D. Hsu, M. Telgarsky. Transformers, parallel computation, and logarithmic depth. *ICML*, 2024.
- A. Stolfo, Y. Belinkov, M. Sachan. A mechanistic interpretation of arithmetic reasoning in language models. *EMNLP*, 2023.
- A. Syed, C. Rager, A. Conmy. Attribution patching outperforms automated circuit discovery. *arXiv:2310.10348*, 2023.
- E. Todd et al. Function vectors in large language models. *ICLR*, 2024.
- W. Wu et al. Retrieval head mechanistically explains long-context factuality. *arXiv:2404.15574*, 2024.
- E. Biran et al. Hopping too late: Exploring the limitations of latent multi-hop reasoning in large language models. *EMNLP*, 2024.
- A. Conmy et al. Towards automated circuit discovery for mechanistic interpretability. *NeurIPS*, 2023.
- M. Dehghani et al. Universal transformers. *ICLR*, 2019.
- M. Elbayad et al. Depth-adaptive transformer. *ICLR*, 2020.
- N. Elhage et al. A mathematical framework for transformer circuits. *Anthropic*, 2021.
- A. Fan, E. Grave, A. Joulin. Reducing transformer depth on demand with structured dropout. *ICLR*, 2020.

- G. Feng et al. Towards revealing the mystery behind chain of thought: A theoretical perspective. *NeurIPS*, 2023.
- J. Ferrando, E. Voita. Information flow routes: Automatically interpreting language models at scale. *EMNLP*, 2024.
- M. Geva et al. Dissecting recall of factual associations in auto-regressive language models. *EMNLP*, 2023.
- A. Graves. Adaptive computation time for recurrent neural networks. *arXiv:1603.08983*, 2016.
- A. Gromov et al. The unreasonable ineffectiveness of the deeper layers. *arXiv:2403.17887*, 2024.
- V. Lad, W. Gurnee, M. Tegmark. The remarkable robustness of LLMs: Stages of inference. *arXiv:2406.19384*, 2024.
- Z. Li et al. Chain of thought empowers transformers to solve inherently serial problems. *ICLR*, 2024.
- X. Men et al. ShortGPT: Layers in large language models are more redundant than you expect. *arXiv:2403.03853*, 2024.
- K. Meng et al. Locating and editing factual associations in GPT. *NeurIPS*, 2022.
- W. Merrill, A. Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *TACL*, 2023.
- P. Michel, O. Levy, G. Neubig. Are sixteen heads really better than one? *NeurIPS*, 2019.
- N. Nanda. Attribution patching: Activation patching at industrial scale. Technical report, 2023.
- nostalgebraist. Interpreting GPT: The logit lens. *LessWrong*, 2020.
- D. Raposo et al. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv:2404.02258*, 2024.
- H. Sajjad et al. On the effect of dropping layers of pre-trained transformer models. *Computer Speech & Language*, 2023.
- T. Schuster et al. Confident adaptive language modeling. *NeurIPS*, 2022.
- E. Voita et al. Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. *ACL*, 2019.
- K. Wang et al. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. *ICLR*, 2023 (arXiv 2022).
- J. Xin et al. DeeBERT: Dynamic early exiting for accelerating BERT inference. *ACL*, 2020.
- S. Yang et al. Do large language models latently perform multi-hop reasoning? *ACL*, 2024.